

3.21 Timers and watchdogs

The devices include two advanced control timers, up to eleven general-purpose timers, two basic timers, two low-power timers, two watchdog timers, and one SysTick timer.

The table below compares the features of the advanced control, general-purpose and basic timers.

Table 7. Timer feature comparison

Timer type	Timer	Counter resolution	Counter type	Prescaler factor	DMA request generation	Capture / compare channels	Complementary outputs
Advanced control	TIM1, TIM8	16 bits	Up, down, Up/down	Any integer between 1 and 65536	Yes	4	4
General-purpose	TIM2, TIM5	32 bits	Up, down, Up/down	Any integer between 1 and 65536	Yes	4	No
	TIM12	16 bits	Up		No	2	No
	TIM15	16 bits	Up, down, Up/down		Yes	2	1
	TIM16, TIM17	16 bits	Up, down, Up/down			1	No
Basic	TIM6, TIM7	16 bits	Up	Any integer between 1 and 65536	Yes	0	No

3.21.1 Advanced-control timers (TIM1/TIM8)

The advanced-control timers (TIM1/TIM8) consist of a 16-bit autoreload counter driven by a programmable prescaler.

They may be used for various purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

TIM1/TIM8 timer features include:

- 16-bit up, down, up/down autoreload counter
- 16-bit programmable prescaler allowing dividing (also “on the fly”) the counter clock frequency by any factor from 1 to 65536
- Seven independent channels for:
 - Input capture (except channels 5, 6, and 7)
 - Output compare
 - PWM generation (edge- and center-aligned mode)
 - One-pulse mode output
- Complementary outputs with programmable dead-time
- Synchronization circuit to control the timer with external signals and to interconnect several timers together
- Repetition counter to update the timer registers only after a given number of cycles of the counter
- Two break inputs to put the timer’s output signals in a safe user-selectable configuration
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization, or count by internal/external trigger)
 - Input capture
 - Output compare
- Incremental encoders, quadrature encoders, and hall-sensors support
- Trigger input for external clock or cycle-by-cycle current management
- ADC synchronization for jitter-free sampling points